



REQUIREMENT DOCUMENT

DoctrNow Teleconsultation Product Requirement Document

“CONFIDENTIAL – Property of DoctrNow”

Table of Contents

1. Overview	6
2. Purpose	6
3. User Roles	7
4. Modules	8
5. Patient Dashboard Requirements	10
5.1. User Authentication & Profile Setup	10
5.2. Home Page	10
5.3. Speciality Selection	11
5.4. Hospital Selection	11
5.5. Doctor Selection	11
5.6. Appointment Booking	12
5.7. Payment Process	12
5.8. Health Details Entry	12
5.9. Consultation Notifications	13
5.10. Virtual Lobby	13
5.11. Consultation	13
5.12. Post-Consultation Process	13
5.13. My Appointments Section	14

5.14. My Prescriptions	15
5.15. My Documents	16
5.16. Follow-Up Messaging	17
5.17. Feedback	17
5.18. Appointment Reminders	17
5.19. Prescription & Medication Adherence	18
5.20. Consent & Privacy Compliance	18
5.21. Customer Support	18
5.22. Compliance & Data Security Requirements	20
6. Doctor Dashboard Requirements	23
6.1. Login Page	23
6.2. Dashboard	23
6.3. Appointments Tab	23
6.4. Appointment Details	24
6.5. Consultation Screen	24
6.6. Prescription Module	25
6.7. Patient Details Panel	26
6.8. Chat Module	26
6.9. Consultation History Tab	26
6.10. System Settings	27

6.11 Consultation Reports	27
7. Hospital Admin Dashboard Requirements	30
7.1. Login Page	30
7.2. Dashboard Page	30
7.3. Consultation Page	32
7.4. Doctors Section	34
7.5. Consultations Section	35
7.6. Patients Section	36
7.7. Revenue Section	38
7.8. Notification Panel & Chat with DoctorNow Admin	41
7.9. Real-Time Payment Update Tracker	42
7.10. Booking Conflict & Resolution Chat	44
7.11. Security & Role-Based Access Control	45
7.12. Scalability & Data Export	47
8. DoctorNow Admin Dashboard Requirement Document	50
8.1. Login Page	50
8.2. Dashboard Page	50
8.3. Hospital Management	51
8.4. Doctor Management	52
8.5. Patient Management	53

8.6. Revenue and Payment Management	53
8.7. Consultation Management	54
8.8. Notification & Communication Management	56
8.9. System & Data Management	56
8.10. Hospital-Specific Admin Assignment	57
8.11. Health Services & Package Management	59
8.12. Data Security & Compliance	60
9. Health Services System Requirements Document	63
9.1. Patient App (Web and Mobile App)	63
9.1.4. Booking & Payment Flow	65
9.1.5. Footer & Navigation	67
9.2. Hospital & Partner Portal (Web)	68
9.3. Admin Dashboard (Web)	70
10. Conclusion	71
11. Feedback Points	72
12. Annexure-Technical Touchpoints	90

DoctorNow Teleconsultation Product Requirement Document

Date: 14 Apr 2025

1. Overview

The DoctorNow Teleconsultation Platform is a comprehensive, cloud-based solution designed to bridge patients, doctors, hospitals, and administrators in a secure, efficient healthcare ecosystem. The platform integrates four core systems for - Patient, Doctor, Hospital, and Admin to enable seamless teleconsultations, appointment management, prescription generation, and real-time communication. Built with HIPAA/GDPR-compliant security protocols, DoctorNow prioritises patient confidentiality while offering intuitive interfaces, automated workflows, and robust analytics. Key features include real-time payment tracking, conflict resolution tools, and scalable support for multi-branch hospital chains.

2. Purpose

The primary purpose of the DoctorNow Platform is to provide patients and healthcare facilities with a robust, user-friendly platform that:

- **Enhance Healthcare Accessibility:** Provide patients with 24/7 access to doctors, health services, and medical records via a unified interface.
- **Streamline Clinical Workflows:** Empower doctors with tools for appointment management, in-session note-taking, prescription generation, and performance tracking.

- **Ensure Data Security:** Encrypt sensitive health data, enforce role-based access and maintain audit logs for compliance.
- **Optimize Hospital Operations:** Enable hospital admins to manage bookings, revenue, staff, and promotions dynamically.
- **Drive Financial Transparency:** Track revenue, commissions, and payments in real-time across doctors, specialties, and branches.
- **Support Scalability:** Handle multi-branch hospital chains with isolated data management and unified admin oversight.
- **Improve Patient Engagement:** Deliver automated reminders, follow-up messaging, and personalized health recommendations.

3. User Roles

1. Patients

- Primary users who book consultations, access medical records, and communicate with doctors.
- Access to personal profiles, family accounts, and appointment scheduling.
- Ability to upload health documents, view prescriptions, and submit feedback.

2. Doctors

- Healthcare providers who conduct teleconsultations and manage patient care.
- Access to appointments, patient profiles, and consultation tools.
- Ability to generate prescriptions, annotate lab reports, and track performance metrics.

3. Hospital Administrators

- Operational managers who oversee hospital workflows, staff, and revenue.
- Access to doctor allocations, consultation schedules, and pricing management.
- Ability to resolve booking conflicts, track revenue, and configure promotional banners.

4. DoctorNow Administrators

- System-wide supervisors ensuring platform compliance and performance.
- Access to hospital and doctor management, commission rates, and audit logs.
- Ability to generate compliance reports, manage payment gateways, and oversee data backups.

4. Modules

Patients - Web and Mobile App

Doctors - Web

Hospital Administrators - Web

DoctorNow Admin - Web

Patient Portal

5. Patient Dashboard Requirements

5.1. User Authentication & Profile Setup

5.1.1. User Authentication

1. The system requires each user to have a unique username and password.
2. The system implements multi-factor authentication for enhanced security.
3. The system enforces password requirements that adhere to healthcare security standards.
4. The system automatically logs out users after a period of inactivity.

Note- Mobile Number is mandatory for Registration

During Registration-OTP to be sent for confirmation.

AUTOPICK OTP into the OTP boxes -by user selection

And post message that registration successful and set

password page.After setting password proceed to profile

setup. Skip option to be provided on top[incase..skip-user

User should navigate through the app and should be able

to see dr,appointmnt,healthservices etc and when doing

booking confirmation. All the details should be shown to

proceed further]

5.1.2. Profile Management

5. The user can complete their profile with the following fields:
 - Personal Details: First name, last name, age, gender, blood group, marital status, Emirates ID ,Nationality ,DOB.
 - Health Information: Existing conditions, allergies, current medications, lifestyle habits.

- The system prompts users to complete an extended health questionnaire post-registration, covering:
- Weight, height, blood pressure, sugar levels (optional).
- Chronic conditions, recent symptoms, and health goals.
- Personal Details: Email id, Emirates ID to be a mandatory field for profile setup
- For different family profile-emirates id should be unique validation to be in place
- Date of birth and emirates id validation to be in place
- Sample-Emirates ID FORMAT -784-1990-941XXXXX.
Here 4 rd to 7th digit is date of birth and first 3 digit is always 784

7.The system prompts users to complete the remaining Health Information:

- Insurance details field to remain as an optional field
- Insurance details: Insurance Company ,InsurancPlan, Insurance Number -Initial release we will keep it as Open Text Fields. Will find api and integrate as dropdown for future release.
- Then followed by Health Information: Existing conditions, allergies, current
- medications, lifestyle habits
- Existing conditions, allergies, current medications, lifestyle habits.
- Weight, height, blood pressure, sugar levels (optional).
- Chronic conditions, recent symptoms, and health goals.

5.2. Home Page

6. Speciality Section: Displays a scrollable list of medical specialities.
7. Hospital Selection Section: Displays preferred hospitals based on location.
8. Existing Patient Section: Displays previously consulted doctors and allows easy follow-up booking.

DoctrNow

9. Search Bar: Allows users to search for doctors, specialities, and facilities via free text input.
11. Home page to have 2 sections one for health service and consultation-hospital listing /speciality listing. Design on how to split the sections or shown both separated in the same home page to be discussed in ui design.
12. Have small horizontal rectangle placeholder for ads -to be controlled or real time updated from hospital admin/super admin

5.3. Speciality Selection

1. The user can view hospitals/clinics based on their location.
2. A location filter is always available on the right side of the interface.
3. On selecting a speciality, the user is directed to a hospital listing page.
4. Additional filters include:
 - Area
 - Speciality

Note : User can view specialty section where specialty selection will be shown as logo of specialty in homepage .User will be able to swipe to right to see all specialty and view all link to be shown to click and view all specialty in detail.

And on selecting that, list of all doctors will be displayed based on the specialty sorted based on locality. On top of the page central search bar with “search Doctor,Speciality,Facility,Location” Filter and sort option to present and user should be able to filter and sort based on

- sort by distance (nearest), or names (a,z)
- and then below Filter Options
1. Consultation date
 2. Doctor Gender
 3. Facility (search and list of center)
 4. Practitioner spoken language
 5. Distance Range
 6. Country of education
 7. Practitioner country of professional Experience
 8. Consultation Fee
 9. Emirate

If patient central search on top “search Doctor, Specialty, Facility, Location”

If user enters doctor name - show the doctor if user enter 3 letters and search, if multiple doctor comes, show multiple doctors

If user enter specialty ,show all doctor related to that specialty

If user enters any facility name-shows all doctors related to that facility

If user enters location name-show all the doctors based on that location of facility

5.4. Hospital Selection

5. The user can view a list of hospitals with the following details:

- Hospital Name
- Location
- Speciality
- Number of Doctors available

6. Displays a list of hospitals based on the selected filters.

7. Hospital section will shown near by 20 hospitals/clinic near by based on location. User can swipe right to see the facility names and view all button to click and see the complete list of hospitals.

8. Hospital List page- sort by distance (nearest) ,or names (a,z) option to be shown

9. On top of the page central search bar with “Search Facility, Location” to be present

- If user selects the hospital/facility/clinic name, the user should be able to see the facility details and below the list of specialty provided by that facility and list of doctors working in that hospital/facility
- The specialty section should be selectable same logo used in homepage, if user in hospital page selects the specialty logo, the doctor list should be filtered and shown.

10. In all the above search results, Each and every doctor tile should have Doctor Name, Doctor Photo ,Doctor speciality ,Doctor Speaking language ,Doctr Clinic/facility name and distance from current patient location ,Next available slot for consultation with date ,time format example, If today or tom (Today,1April at 6.30PM).else (10th April at 6.30PM) and view more option

and a tag option to save doctor to my save list for easy reference.

11 .Once click on doctor tile-details as per excel

- Doctor Photo with name
- Specialty
- Language spoken
- Show available section which shows latest date available slots with option to view more slot
- Doctor details
- Doctor Qualification
- Doctor specialty
- Doctor services

Selected patient profile with option to change in case of multiple family profile

Example-Kids to be seen by pediatric doctors alone .If kids selected for adult doctor -validation to be shown dob >18 to be allowed message to be shown

And adults cannot be seen by pediatric doctor

12.All the details to be populated based on entry from Hospital Admin portal or App admin portal and if new doctor added should be reflected immediately in real time or max of 30min.Addition of doctor should have maker checker process

5.5. Doctor Selection

11.Upon selecting a hospital, the system lists all available doctors at that hospital with the following details:

- Name
- Specialty
- Qualifications
- Available Time Slots

12.Users can search and filter the list.

13.Displays real-time availability or estimated wait time for doctors.

14. The user is an existing customer, this section also shows a list of previously consulted doctors.

5.6. Appointment Booking

15. The user selects an available time slot for consultation.

16. A summary page displays the appointment details before confirmation.

17. The user proceeds to the payment page.

5.7. Payment Process

18. Displays a detailed payment summary.

19. The user can:

- Pay immediately via card.
- Choose 'Pay Later' (payment must be completed at least 30 minutes before the consultation).
- As a patient, I can choose 'Pay with Insurance' when booking an appointment that is scheduled at least 6 hours in advance. I enter my Emirates ID and insurance number (with format validation). Upon clicking 'Request Insurance Co-Payment', the appointment slot is temporarily blocked with status 'Pending Insurance Verification'.

1. A unique transaction ID is generated and this booking cannot be modified or edited further by the user. The only option now is to cancel and rebook or switch to full payment if needed.

2. In the hospital admin module, under 'Insurance Verification', the hospital staff can view the booking details and manually verify the patient's eligibility through the EHR system. If eligible, they input the co-payment amount and click 'Request Payment'. The patient receives a push notification in real-time stating: 'Your insurance has been approved. Please pay AED XX as co-

payment to confirm your appointment.'

3. Once the payment is made by the patient, the system updates the appointment status to 'Confirmed'. All steps are logged and protected from tampering. If the patient chooses not to pay, the slot remains blocked for up to 4 hours, after which it expires unless converted to full payment manually.

20. Payment reminders are sent via push notifications:

- 1 hour before
- 45 minutes before
- 30 minutes before

21. If payment is not made by the 30-minute mark, a pop-up asks if the appointment should be cancelled or rescheduled.

22. Upon successful payment, a confirmation SMS is sent.

5.8. Health Details Entry

23. Page 1: The user can enter the following details: Weight, Height, BP, and Sugar Level, these details are optional.

24. Page 2: Option to describe consultation reason and upload up to 5 medical documents.

25. Uploaded documents are securely stored in the cloud and require OTP verification for access.

26. If skipped, details can be entered later via the "My Appointments" tab.

5.9. Consultation Notifications

33. 15-minute reminder: "Your doctor will join in 15 minutes. Please join on time."

34. 3-minute reminder: "Your doctor will join in 3 minutes. Please join the virtual lobby."

5.10. Virtual Lobby

35. Users enter a waiting area similar to Google Meet until the doctor joins.

36. Once the doctor joins, a notification appears, and the video call begins.

5.11. Consultation

37. Once the Doctor Joins, the User will get a message "Doctor has joined, and video consultation will begin now" with a Note "Call will be analysed with secured algorithms for foul language"

38. During the consultation, the patients can:

- Send text messages to the doctor.
- Share medical reports from their mobile device.
- Capture and send images using the mobile camera.

5.12. Post-Consultation Process

39. A notification confirms consultation completion.

40. If a prescription is provided, it appears under "My Appointments -> Today's Consultations" within 10 minutes.

41. Users can email the prescription for medicine purchase.

5.13. My Appointments Section

5.13.1. Appointment Overview

42. The user can view a list of all their appointments, categorized into:

- Upcoming Appointments: Scheduled consultations with date, time, doctor name, and status (e.g., confirmed, pending payment).
- Past Appointments: Completed consultations with date, doctor name, and prescription/report availability.

43. The system displays a summary card for each appointment, including:

- Doctor's name, speciality, and profile picture.
- Appointment date, time, and duration.
- Status (e.g., confirmed, cancelled, rescheduled).

5.13.2. Appointment Management

44. The user can reschedule or cancel upcoming appointments, subject to the platform's cancellation policy.

45. The system sends confirmation notifications for any changes made to the appointment.

46. The user can join upcoming consultations directly from the appointment card by clicking a "Join Now" button.

5.13.3. Prescription & Report Access

47. The user can view and download prescriptions, consultation notes, and lab reports associated with past appointments.

48. The system allows users to email prescriptions or reports directly from the appointment details page.

5.13.4. Health Details & Document Upload

49. The user can update health details (e.g., weight, and blood pressure) or upload documents (e.g., lab reports) for upcoming appointments.
50. The system ensures all uploaded documents are securely stored and accessible only to the user and the assigned doctor.

5.14. My Prescriptions

5.14.1. Prescription List

51. The system displays a list of all prescriptions issued to the patient, including:
- Prescription Date: The date the prescription was issued.
 - Doctor Name: Name of the doctor who issued the prescription.
 - Medications: List of prescribed medications with dosage instructions.
 - Health Recommendations: Additional advice or follow-up actions are provided by the doctor.
52. The user can Download the list in PDF/CSV/XLSX.

5.14.2. Prescription Details

53. The patient can click on a prescription to view detailed information, including:
- Medication Details: Name, dosage, frequency, and duration.
 - Doctor Notes: Any additional notes or instructions from the doctor.
 - Validity Period: Expiry date of the prescription (if applicable).
54. The patient can download prescriptions in PDF format for offline use.
55. The system allows patients to share prescriptions via email or other communication channels.

5.15. My Documents

5.15.1. Document List

56. The system displays a list of all documents uploaded by the hospital or patient, including:

- Document Type: Lab reports, test results, referral letters, discharge summaries, etc.
- Upload Date.
- Associated Consultation.
- Document Title.
- Description.
- Download & Share options.
- Associated Consultation: Consultation date and doctor name (if applicable).

57. The user can Download the list in PDF/CSV/XLSX.

5.15.2. Document Details

58. The patient can click on a document to view detailed information, including:

- Document Title: Name of the document (e.g., "Blood Test Report").
- Description: Summary of the document's content.
- Tags: Patient name and date for easy identification.

59. The patient can download documents in their original format (e.g., PDF, JPEG).

60. The system allows patients to share documents via email or other communication channels.

5.15.3. Search & Filter

61. The patient can search for specific prescriptions or documents using:

- Keywords: Search by medication name, document title, or doctor name.
- Date Range: Filter by a specific period.
- Document Type: Filter by lab reports, prescriptions, etc.

5.15.4. Notifications

62. The system sends notifications to the patient when:

- A new prescription is issued.
- A new document is uploaded.

63. The patient can configure notification preferences (e.g., email, push notifications).

5.16. Follow-Up Messaging

64. Users can send up to 5 messages within 4 days after the consultation via the "Past Appointments" section.

65. The chat interface allows text-based communication.

5.17. Feedback

66. Users can submit feedback within 24 hours post-consultation.

67. After 24 hours, the appointment moves to "Past Appointments," and the feedback link disappears.

5.18. Appointment Reminders

68. Push notifications and SMS reminders are sent:

- 24 hours before the appointment.
- 1 hour before the appointment.

69. Users can add appointments to their device calendars.

5.19. Prescription & Medication Adherence

- 70. The system sends automated reminders to users to take prescribed medications at scheduled times.
- 71. The user can set custom reminder schedules (e.g., morning, afternoon, evening) for each medication.
- 72. The system sends reminders via push notifications, SMS, or email based on user preference.

5.20. Consent & Privacy Compliance

- 73. Before uploading personal health information or joining a consultation, users must agree to a digital consent form.
- 74. Data security follows HIPAA/GDPR compliance standards.

5.21. Customer Support

5.21.1 24/7 Live Chat Support

- 75. The system provides a live chat feature accessible from the Patient Dashboard for immediate technical support.
- 76. The user can initiate a chat session by clicking a "Support" button prominently displayed on the dashboard.
- 77. The system ensures live chat is available 24/7, with support agents ready to assist with:
 - Technical issues (e.g., video call failures, payment gateway errors).
 - Account-related queries (e.g., password reset, profile updates).
 - General platform navigation assistance.
- 78. The system displays estimated wait times for chat responses to manage user expectations.
- 79. The user can attach screenshots or error messages during the chat to help resolve issues faster.

80. The system provides a transcript of the chat session, which the user can download or email for future reference.

5.21.2 Hotline Support

81. The system provides a toll-free hotline number for users who prefer phone support.

82. The user can access the hotline number from the "Support" section of the dashboard.

83. The system ensures the hotline is operational 24/7, with multilingual support agents (if applicable).

84. The system routes calls to the appropriate support team based on the issue type (e.g., technical, billing, account).

85. The user can request a callback if wait times exceed a predefined threshold (e.g., 5 minutes).

5.21.3 Support Ticket System

86. The user can submit a support ticket for non-urgent issues via the "Support" section.

87. The system allows users to categorize their tickets (e.g., technical, billing, feedback).

88. The user can track the status of their support tickets in real time, with updates sent via email or push notifications.

89. The system ensures support tickets are resolved within a predefined SLA (e.g., 24 hours for non-critical issues).

5.21.4 Self-Help Resources

90. The system provides a comprehensive FAQ section accessible from the "Support" menu.

91. The user can search for solutions using keywords or browse topics (e.g., "How to book an appointment," "Troubleshooting video calls").

- 92. The system includes step-by-step guides and video tutorials for common tasks.
- 93. The user can access troubleshooting tools (e.g., network diagnostics, and browser compatibility checks) from the support section.

5.21.5 Integration with Other Modules

- 94. The system integrates customer support with the appointment and payment modules to provide context-aware assistance.
- 95. For example, if a user reports a payment failure, the support agent can view the transaction details in real-time.
- 96. The system allows support agents to initiate video calls with users for complex technical issues.

5.22. Compliance & Data Security Requirements

5.22.1. HIPAA & GDPR Compliance

- 97. Ensure end-to-end encryption for data in transit and at rest to protect patient confidentiality.
- 98. Obtain explicit user consent before collecting, processing, or storing personal health data, in line with HIPAA and GDPR.
- 99. Implement secure cloud infrastructure with failover support to ensure high availability and data protection.
- 100. Restrict access to sensitive data based on role-based access control (RBAC) to minimize unauthorized access.
- 101. Ensure compliance with GDPR's "Right to be Forgotten" by allowing users to request data deletion securely.
- 102. Implement audit logs to track access and modifications to patient records for compliance verification.

5.22.2. Data Security & Privacy

- 103. Encrypt patient data both in storage and transmission to prevent unauthorized access.
- 104. Store health records in a secure cloud environment with regular security audits.
- 105. Require multi-factor authentication (MFA) for system access by healthcare providers.
- 106. Enable automatic session timeouts to prevent unauthorized use on unattended devices.

Doctor Dashboard

6. Doctor Dashboard Requirements

6.1. Login Page

1. The doctor can securely log in using a password and/or facial authentication.
2. The system should ensure quick and seamless access from both desktop and laptop devices.
3. The system should support multi-factor authentication for added security.

6.2. Dashboard

4. The doctor can view a daily summary of scheduled appointments, notifications, and key performance metrics.
5. The system should display:
 - The number of consultations for today and the next seven days.
 - Pending notifications and alerts.
 - Key performance indicators (e.g., consultation counts, pending feedback).
6. The system should allow filtering data by daily, weekly, and monthly views.

6.3. Appointments Tab

7. The doctor can view a list of scheduled appointments for today and the next seven days.
8. The system should display:
 - Patient names, appointment times, and brief patient details.
 - Appointment status (upcoming, completed, cancelled).
9. The system should allow sorting and filtering based on patient name, appointment date, and status.

6.4. Appointment Details

6.4.1. Appointment Acceptance

10. The doctor can receive a pop-up notification for incoming consultation requests.
11. The system should confirm appointments and add them to the doctor's active consultations list.
12. The system should prevent the dismissal of the confirmation pop-up without accepting or rejecting the request.

6.4.2. Pre-Consultation Reminders

13. The doctor can receive automatic reminder pop-ups at:
 - 5 minutes before the scheduled consultation.
 - 1 minute before the scheduled consultation.
 - 15 seconds before the scheduled consultation.
14. The system should not allow the 15-second pop-up to be dismissed; the doctor must accept and join the call.

6.5. Consultation Screen

6.5.1. Consultation Timer

15. The system should automatically start a timer when a consultation begins.
16. The doctor can track session duration in real time.

6.5.2. Extend Consultation Time

17. The doctor can extend the consultation by one or two minutes if needed.
18. The system should display a confirmation pop-up before extending.
19. The system should check upcoming bookings before allowing an extension.

6.5.3. Cancellation Prompt

- 20. The doctor can receive a cancellation prompt if the patient does not join on time.
- 21. The system should ask for confirmation before cancelling the booking.
- 22. The system should ensure the booking amount is refunded and a no-show fee is charged to the patient.

6.5.4. In-Session Note Taking

- 23. The doctor can type and save notes during the consultation.
- 24. The system should allow retrieval and modification of notes after the consultation.

6.6. Prescription Module

6.6.1. Prescription Generation

- 25. The doctor can select medications from an up-to-date list (covering all medications available in the UAE).
- 26. The system should generate a prescription automatically with dosage details.
- 27. The system should send the prescription to the patient via email and display it in their portal.

6.6.2. Adding Health Recommendations

- 28. The doctor can suggest additional health tests or check-ups alongside prescriptions.
- 29. The system should display recommendations in a clear, formatted list for patient reference.

6.7. Patient Details Panel

6.7.1. Patient Profile & History Access

30. The doctor can access a detailed patient profile with:

- Personal details.
- Previous consultation history.
- Past lab reports.

6.7.2. Lab Report Review & Annotation

31. The doctor can view, filter, and annotate lab reports.

32. The system should allow filtering by date and type.

33. The system should support adding notes or annotations to reports.

6.8. Chat Module

34. The doctor can send and receive messages from patients during and outside consultations.

35. The system should:

- Support up to five messages per conversation.
- Send pop-up notifications for new messages when the doctor is not in a session.

6.9. Consultation History Tab

36. The doctor can view past consultations along with associated notes and prescriptions.

37. The system should display:

- A chronological list of past consultations.
- Search and filter options based on patient name or consultation date.

6.10. System Settings

38. The doctor can configure notification preferences for desktop usage.

39. The system should allow customization of:

- Sound alerts.
- Pop-up notifications.
- Email alerts.

6.11 Consultation Reports

6.11.1 Consultation Summary

40. The system displays the total number of consultations for the current month, categorized as:

- Booked: Appointments scheduled but not yet completed.
- Completed: Appointments that have been successfully conducted.
- Upcoming: Appointments scheduled for future dates.

41. The system provides a visual representation (e.g., bar chart, pie chart) of the consultation breakdown for quick insights.

6.11.2 Filter Options

42. The doctor can filter consultation data by:

- Date Range: View data for a specific day, week, or custom date range.
- Status: Filter by booked, completed, or upcoming consultations.
- Patient Name: Search for consultations involving a specific patient.

43. The system updates the displayed data dynamically based on the selected filters.

6.11.3 Detailed Consultation List

44. The system provides a detailed list of consultations, including:

- Patient Name: Name of the patient for each consultation.
- Appointment Date & Time: Scheduled date and time of the consultation.
- Status: Current status (booked, completed, cancelled).
- Duration: Length of the consultation (for completed sessions).
- Feedback: Patient feedback and ratings (if available).

45. The doctor can sort the list by:

- Appointment date (ascending/descending).
- Patient name (alphabetical order).
- Consultation status.

6.11.4 Export Functionality

46. The doctor can export consultation reports in multiple formats, including:

- CSV: For data analysis in spreadsheet software.
- PDF: For sharing or printing.

Hospital Administrator

7. Hospital Admin Dashboard Requirements

7.1. Login Page

47. The hospital admin enters their email and password on the login page.
48. On successful login, the system redirects the admin to the Dashboard page.
49. If the credentials are invalid, an inline error message appears: "Invalid credentials. Please try again."
50. The system enforces secure login standards, including password complexity and potential 2FA in future updates.
51. The login page contains fields for email and password, along with a login button.

7.2. Dashboard Page

7.2.1. View Dashboard Welcome Message

52. The system should display a welcome message such as:
"Welcome, [Admin Name]!" along with the hospital name.
53. The dashboard header includes navigation links to:
 - Dashboard
 - Doctors
 - Consultations
 - Revenue
54. The main section displays an overview of key hospital statistics, including:
 - Number of Doctors Available
 - Number of Teleconsultations Completed
 - Number of Specialties Served
 - Revenue Generated

7.2.2. View Total Consultations on the Dashboard

55. The system displays a summary widget on the dashboard showing the total number of consultations for the current month, categorized as:

- Booked: Appointments scheduled but not yet completed.
- Completed: Appointments that have been successfully conducted.
- Upcoming: Appointments scheduled for future dates.

56. The widget provides a quick visual representation (e.g., bar chart, pie chart) of the consultation breakdown.

57. The hospital admin can click the widget to be redirected to the detailed Consultation Page.

7.2.3. View Revenue Summary on Dashboard

58. The system displays a revenue summary widget showing revenue for the past year, 6 months, and current month.

59. Clicking the widget redirects to the Revenue Page, where the admin can:

- View revenue generated per doctor and speciality.
- See commission details shared with DoctorNow for different timeframes.

7.3. Consultation Page

7.3.1. Consultation Page Overview

60. The system displays a detailed list of consultations on the Consultation Page, including:

- Patient Name: Name of the patient for each consultation.
- Appointment Date & Time: Scheduled date and time of the consultation.
- Status: Current status (booked, completed, cancelled).
- Doctor Name: Name of the doctor assigned to the consultation.
- Specialization: Doctor's specialization for the consultation.
- Duration: Length of the consultation (for completed sessions).
- Feedback: Patient feedback and ratings (if available).

7.3.2. Filter Options

61. The hospital admin can filter consultations by:

- Date Range: View data for a specific day, week, or custom date range.
- Status: Filter by booked, completed, or cancelled consultations.
- Doctor Name: Search for consultations involving a specific doctor.
- Patient Name: Search for consultations involving a specific patient.

62. The system updates the displayed data dynamically based on the selected filters.

7.3.3. Sorting Options

63. The hospital admin can sort the consultation list by:

- Appointment Date: Ascending or descending order.
- Patient Name: Alphabetical order.
- Doctor Name: Alphabetical order.
- Status: Group consultations by status (booked, completed, cancelled).

7.3.4. Detailed Consultation View

64. The hospital admin can click on any consultation entry to view detailed information, including:

- Patient Profile: Name, age, gender, contact details.
- Medical History: Past consultations, prescriptions, and lab reports.
- Consultation Notes: Notes taken by the doctor during the session.
- Prescription Details: Medications prescribed, dosage, and health recommendations.

65. Feedback: Patient feedback and ratings (if available).

7.3.5. Export Functionality

66. The hospital admin can export the consultation data in multiple formats, including:

- CSV: For data analysis in spreadsheet software.
- PDF: For sharing or printing.

67. The system ensures that exported reports include all filtered and sorted data.

7.4. Doctors Section

7.4.1. View Doctor Allocation List

68. The hospital admin can view a table displaying all doctors with:

- Name
- Specialization
- Availability
- Specialty
- License Number
- About Doctor
- Qualification
- Consultation Charges

69. Each doctor's row contains an "Edit" button to modify details.

70. The hospital admin can update consultation charges but not other details.

7.4.2. Add a New Doctor

71. The hospital admin can click "Add Doctor," filling out a form including:

- Name
- Specialization
- Availability
- Specialty
- License Number
- About Doctor
- Qualification
- Consultation Charges

72. After submission, the system displays a success message.

7.4.3. Edit Doctor Details

73. The hospital admin can click "Edit" on a doctor's row to modify:

- Name
- Specialization
- Availability
- About Doctor
- Qualification
- Consultation Charges

74. Changes are reflected immediately in the doctor allocation list.

7.4.4. Remove a Doctor

75. The hospital admin can click "Remove" next to a doctor.

76. A confirmation popup appears before deletion.

77. Once confirmed, the doctor is removed from the list.

7.5. Consultations Section

78. The hospital admin can view a list of consultations with:

- Patient Name
- Doctor
- Specialization
- Date & Time

79. The hospital admin can select a consultation to update its time or date.

80. Upon confirmation, notifications are sent to both the patient and the doctor.

81. The updated schedule appears in both dashboards.

82. The hospital admin can filter appointments by date, patient name, or consultation type.

83. The system can highlight urgent or rescheduled appointments.

7.6. Patients Section

7.6.1. Patient List

84. The system displays a list of all patients associated with the hospital, including:

- Patient Name: Full name of the patient.
- Contact Information: Phone number and email address.
- Registration Date: Date the patient registered with the hospital.
- Last Consultation Date: Date of the patient's most recent consultation.

85. The system allows the admin to search for patients by:

- Name: Full or partial name.
- Contact Information: Phone number or email.
- Registration Date: Filter by a specific date range.

7.6.2. Patient Profile

86. The hospital admin can click on a patient's name to view their detailed profile, including:

- Personal Information: Name, age, gender, blood group, and marital status.
- Contact Details: Phone number, email, and address.
- Medical History: Past consultations, diagnoses, and treatments.
- Uploaded Documents: Lab reports, prescriptions, and other medical records.

87. The system ensures that sensitive medical data is accessible only to authorized roles (e.g., doctors, and records staff).

7.6.3. Consultation History

88. The system displays a chronological list of the patient's consultations, including:

- Date & Time: When the consultation occurred.
- Doctor Name: Name of the doctor who conducted the consultation.
- Specialization: Doctor's specialization for the consultation.
- Notes: Summary of the consultation and recommendations.
- Prescriptions: Medications prescribed during the consultation.

89. The hospital admin can filter the consultation history by:

- Date Range: View consultations within a specific period.
- Doctor Name: Filter by a specific doctor.
- Specialization: Filter by a specific medical speciality.

7.6.4. Lab Reports & Documents

90. The system provides access to all lab reports and documents uploaded by the patient, including:

- Lab Reports: Blood tests, imaging reports, etc.
- Prescriptions: Medications prescribed by doctors.
- Other Documents: Referrals, discharge summaries, etc.

91. The system allows the hospital admin to filter documents by:

- Document Type: Lab reports, prescriptions, etc.
- Date: View documents from a specific date range.

7.6.5. Patient Communication

92. The system allows the admin to view past communication with the patient, including:

- Messages: Chat history between the patient and doctors.
- Notifications: Appointment reminders, payment alerts, etc.

93. The system ensures that communication logs are stored securely and accessible only to authorized roles.

7.6.6. Export Patient Data

94. The hospital admin can export patient data in CSV or PDF format, including:

- Patient Profile: Personal and contact information.
- Consultation History: List of consultations with details.
- Lab Reports & Documents: List of uploaded documents.

95. The system ensures that exported data is anonymized for compliance with data protection regulations.

7.7. Revenue Section

7.7.1. Revenue Page Overview

96. The system displays a detailed breakdown of revenue on the Revenue Page, including:

- Total Revenue: Overall revenue generated for the selected timeframe.
- Revenue by Doctor: Revenue generated by each doctor for the selected timeframe.
- Revenue by Specialty: Revenue generated by each speciality for the selected timeframe.
- Commission Details: The Commission shared with the platform (e.g., DoctorNow) for the selected timeframe.

7.7.2. Filter Options

97. The hospital admin can filter revenue data by:

- Timeframe: View data for the past year, last 6 months, last month, current month, or a custom date range.
- Doctor: Filter revenue generated by a specific doctor.
- Speciality: Filter revenue generated by a specific speciality.

98. The system updates the displayed data dynamically based on the selected filters.

7.7.3. Sorting Options

99. The hospital admin can sort the revenue data by:

- Revenue Amount: Ascending or descending order.
- Doctor Name: Alphabetical order.
- Speciality: Alphabetical order.
- Timeframe: Group revenue data by selected timeframe.

7.7.4. Detailed Revenue View

100. The hospital admin can click on any revenue entry to view a list with the following details:

- Doctor Name: Name of the doctor generating the revenue.
- Specialty: Specialty associated with the revenue.
- Consultation Count: Number of consultations contributing to the revenue.
- Commission: Total commission shared with the doctor.
- Net Revenue: Revenue after deducting the platform's commission.

101. The hospital admin can search the list using the Doctor's name.

102. The hospital admin can filter the list by Speciality and Date Range.

7.7.5. Export Functionality

103. The hospital admin can export the revenue data in multiple formats, including:

- CSV: For data analysis in spreadsheet software.
- PDF: For sharing or printing.

104. The system ensures that exported reports include all filtered and sorted data.

7.7.6. View & Manage Commission Details

105. The system displays the agreed commission rate with the telehealth platform.
106. The hospital admin can view the revenue split (hospital vs. platform share) in real-time.
107. The hospital admin can filter by:
- Date Range
 - Doctor
 - Department

7.7.7. Attach Reimbursement Form

108. The hospital admin can upload or generate a reimbursement form (PDF or e-form) within 15 minutes post-consultation.
109. The form appears in the patient's portal for download.
110. The hospital admin can track form status (viewed/unviewed by the patient).

7.7.8. Price Modification

111. The hospital admin can set consultation and service charges for each doctor.
112. The system allows the admin to modify charges for:
- Consultation Fees: Charges for teleconsultations.
 - Service Fees: Charges for additional services (e.g., health check-ups, tests).
113. The system displays the current charges for each doctor and service with the following details:
- Doctor Name
 - Specialty
 - Consultation Rate
114. The hospital admin can confirm modifications, after which the updated charges are reflected immediately in the app.

115. The system ensures that the updated charges are visible to patients during booking.

116. The system maintains an audit log of all price modifications, including:

- Modified By: Admin who made the change.
- Date & Time: When the change was made.
- Previous & Updated Charges: Details of the modification.

7.8. Notification Panel & Chat with DoctorNow Admin

7.8.1 Notification Panel

117. The system displays system alerts in a dedicated notification panel, including:

- Payment Updates: Notifications for successful, failed, or pending payments.
- Schedule Conflicts: Alerts for overlapping bookings or resource conflicts.
- New Feature Announcements: Updates about new platform features or enhancements.

118. The system categorizes notifications by type (e.g., payment, scheduling, announcements) for easy filtering.

119. The hospital admin can mark notifications as read or unread and delete them as needed.

7.8.2 Chat with DoctorNow Admin

120. The hospital admin can initiate a chat with the platform's central admin for issue resolution by entering the following details:

- Subject
- Issue Type (Drop Down)
- Description
- Attachments

121. The system supports real-time messaging with features such as:

- Typing indicators.
- Read receipts.
- File attachments (e.g., screenshots, documents).

122. The system stores all chat conversations for future reference.

7.8.3 Search & Filter

123. The hospital admin can search past chat conversations by:

- Date range.
- Issue type (e.g., payment, scheduling).

124. The system allows filtering of chat history by resolved and unresolved issues.

7.8.4 Audit Logs

125. The system maintains an audit log of all notifications and chat interactions, including:

- Notification Details: Type, date, and status (read/unread).
- Chat History: Participants, date, and conversation summary.

7.9. Real-Time Payment Update Tracker

7.9.1 Live Payment Dashboard

126. The system displays a live dashboard showing ongoing transactions for:

- Consultations: Payments for teleconsultations.
- Test Packages: Payments for health check-ups or lab tests.
- Other Services: Payments for additional services (e.g., nurse-at-home).

127. Each list has the following details:

- Transaction Type (consultation, test package, other services).
- Patient Name.

- Amount.
- Status (pending, successful, failed).
- Date & Time.

128. The system categorises transactions by status:

- Pending: Payments awaiting confirmation.
- Successful: Payments successfully processed.
- Failed: Payments that could not be completed.

129. The hospital admin can filter the list by status and date range.

7.9.2 Transaction Details

130. The hospital admin can drill down into each transaction to view details, including:

- Date & Time: When the transaction occurred.
- Patient Name: Name of the patient associated with the transaction.
- Amount: Total payment amount.
- Service Type: Consultation, test package, or other service.

131. The system updates transaction statuses in real-time.

7.9.3 Partial Payments & Refunds

132. The system automatically updates the dashboard for:

- Partial Payments: Transactions where only a portion of the amount is paid.
- Refunds: Transactions where payments are refunded to the patient.

133. The system sends notifications to both the admin and the patient for:

- Successful payments.
- Failed payments.
- Partial payments.

- Refunds.

7.10. Booking Conflict & Resolution Chat

7.10.1 Conflict Detection

134. The system automatically detects and flags overlapping bookings or resource conflicts, such as:

- Double-booked doctors.
- Overlapping consultation times.
- Resource unavailability (e.g., equipment, rooms).

135. The system displays flagged conflicts in a dedicated Conflict Resolution section with the following details:

- Patient name
- Email
- Consulting Doctor
- Conflict type (Double-booked doctors, Overlapping consultation times, Resource unavailability (e.g., equipment, rooms).)

136. The hospital admin can select a patient to provide a resolution.

7.10.2 Conflict Resolution

137. The hospital admin can resolve conflicts by:

- Reassigning Patients: Assigning the patient to another available doctor.
- Rescheduling Consultations: Moving the consultation to a different time or date.

138. The system provides real-time availability updates to assist in conflict resolution.

7.10.3 Chat with Central Admin

139. The hospital admin can initiate a chat with the platform's central admin to discuss conflict resolution options.

140. The system supports real-time messaging with features such as:

- Typing indicators.
- Read receipts.
- File attachments (e.g., screenshots, schedules).

141. The system stores all chat conversations for future reference.

7.10.4. Security & Role-Based Access Control

142. The hospital admin can create sub-accounts with restricted access based on roles:

- Finance (Revenue View Only)
- Scheduling (Consultations Only)
- Records (Limited Patient Data)

143. All actions (add/edit/remove) are logged for compliance.

7.11. Security & Role-Based Access Control

7.11.1 Sub-Account Creation

144. The hospital admin can create sub-accounts for different departments or staff roles, including:

- Finance: Access is limited to revenue data only.
- Scheduling: Access is limited to consultation scheduling only.
- Records: Access is limited to patient data (excluding sensitive medical information).

145. The system allows the admin to define granular permissions for each role, ensuring restricted access based on job responsibilities.

7.11.2 Secure Login Standards

146. The system enforces secure login standards for all sub-accounts, including:

- Password Complexity: Minimum length, special characters, and alphanumeric requirements.
- Two-Factor Authentication (2FA): Optional but recommended for added security.

147. The system automatically logs out users after a period of inactivity.

7.11.3 Role-Based Access

148. The system restricts access to sensitive information based on user roles:

- Finance Role: Can view revenue reports but cannot access patient medical data.
- Scheduling Role: Can manage consultations but cannot view financial data.
- Records Role: Can access limited patient data (e.g., appointment history) but cannot view revenue or sensitive medical records.

7.11.4 Action Logging

149. The system logs all actions performed by sub-account users, including:

- Add: Logs when a new entry is created (e.g., adding a doctor or consultation).
- Edit: Logs changes made to existing entries (e.g., updating consultation details).
- Remove: Logs deletions (e.g., removing a doctor or consultation).

150. The system stores logs with the following details:

- User: Name and role of the user acting.
- Date & Time: When the action was performed.

- Action Type: Add, edit, or remove.

151. Details: Specific changes made (e.g., old and new values).

7.11.5 Audit & Compliance

152. The system ensures all logs are accessible for auditing and compliance purposes.

153. The system generates compliance reports, including:

- Access logs.
- Action logs (add/edit/remove).
- Security incidents (e.g., failed login attempts).

7.11.6. System Alerts & Notification Buffer

154. If the system was offline or the admin was logged out, missed notifications are buffered.

155. System-wide alerts (maintenance, updates) appear in a banner.

156. The hospital admin can snooze or dismiss them, but they remain in a "History" tab.

7.12. Scalability & Data Export

7.12.1 Scalability for Multiple Branches

157. The system supports hospital chains with multiple branches, allowing data to be managed separately for each branch.

158. The system provides a unified view for admins who oversee multiple branches, enabling them to:

- Access data for all branches under a single login.
- Switch between branches seamlessly.
- The system ensures data isolation between branches to maintain privacy and compliance.

7.12.2 Data Export Functionality

159. The system allows admins to export data in the following formats:

- CSV: For data analysis in spreadsheet software.
- PDF: For sharing or printing.

160. The system supports exporting data for:

- Monthly Reports: Data for the current or past month.
- Quarterly Reports: Data for the current or past quarter.
- Annual Reports: Data for the current or past year.

161. The system ensures that exported reports include all filtered and sorted data.

7.12.3 Report Types

162. The system provides pre-configured templates for common reports, such as:

- Doctor Reports: Doctors Data
- Patient Reports: Patient Data
- Financial Reports: Revenue, commissions, and expenses.
- Compliance Reports: Audit logs, access logs, and action logs.
- Operational Reports: Consultation statistics, doctor performance, and patient feedback.

163. The admin can customize reports by selecting specific data fields and filters.

DoctorNow Admin Dashboard

8. DoctorNow Admin Dashboard Requirement Document

8.1. Login Page

1. The TeleHealthAdmin enters their email and password on the login page.
2. On successful login, the system redirects the admin to the Dashboard page.
3. If the credentials are invalid, an inline error message appears: "Invalid credentials. Please try again."
4. The system enforces secure login standards, including password complexity and potential 2FA in future updates.
5. The login page contains fields for email and password, along with a login button.

8.2. Dashboard Page

6. The system displays an overview of key TeleHealth statistics, including:
 - Number of Consultations: Booked, Completed, Upcoming
 - Revenue Generated
 - App Commission
7. The dashboard provides filter options to view data by:
 - Hospital
 - Doctor
 - Patient

8.3. Hospital Management

8.3.1. View List of Hospitals

8. The admin can view all hospitals with the following details:

- Name
- Location
- Specialties
- Total Consultations

9. Each hospital entry has options to:

- Edit hospital details
- View associated doctors

8.3.2. Add New Hospital

10. The admin can add a hospital by entering the following details:

- Name
- Location
- Specialties
- Doctors

11. Upon successful addition, a confirmation message is displayed.

8.3.3. Edit Hospital Details

12. The admin can edit hospital details such as:

- Location
- Specialties
- Associated doctors

13. Changes go through the maker-checker approval process.

8.3.4. Approve/Reject Hospital Updates

14. The checker can view pending hospital updates and either approve or reject them.
15. Approved changes are applied to the hospital profile.

8.4. Doctor Management

8.4.1. View List of Doctors

16. The admin can view all doctors with details:
 - Name
 - Specialization
 - Availability
 - Consultations
17. Each doctor entry has options to:
 - View profile
 - Edit details
 - Deactivate account

8.4.2. Add New Doctor to Hospital

18. The admin can add a new doctor to a hospital by specifying:
 - Name
 - Specialization
 - Availability
 - Payout structure

8.4.3. Deactivate a Doctor

19. The admin can deactivate a doctor's profile after confirmation.
20. A notification is sent to the doctor and the associated hospital.

8.5. Patient Management

8.5.1. View Patient List

21. The admin can view all patients with details:

- Name
- Contact information
- Consultation history

22. Clicking a patient's name opens their detailed profile.

8.5.2. View Patient Details

23. The system displays the patient's profile, including:

- Biodata
- Medical history
- Consultation history
- Uploaded reports

8.6. Revenue and Payment Management

8.6.1. View Revenue Reports

24. The admin can view detailed revenue reports filtered by:

- Hospital
- Doctor
- Date Range

25. Reports include:

- Total consultations
- App commissions
- Payouts

8.6.2. Commission Rate Management

26. The admin can configure or adjust commission rates per:

- Hospital
- Doctor
- Health service

27. The system reflects changes in real-time across patient payments and revenue reports.

8.6.3. Payment Gateway & Refund Settings

28. The admin can integrate new payment gateways (e.g., PayPal, Stripe, local banks)

29. The admin can manage refund policies (partial or full refunds)

30. The admin can set up advanced rules (e.g., auto-refund for cancellations before 24 hours)

31. The system logs all payment/refund activities for auditing.

32. The system also maintains an audit log of who changed what and when.

8.7. Consultation Management

8.7.1. View Consultation Logs

33. The admin can view logs of all consultations, including:

- Patient
- Doctor
- Hospital
- Date
- Status

34. The logs are filterable by:

- Date Range
- Doctor
- Hospital

8.7.2. Advanced Appointment Overview

35. The admin can access a detailed breakdown of appointments

filtered by:

- Day, Week, or Month
- Hospital
- Doctor Specialty
- Insurance Coverage

36. The system highlights:

- No-shows
- Cancellations
- Overbookings

37. The admin can manually reschedule appointments or override conflicts with documented reasons.

8.7.3. Appointment Conflict & Escalation Management

38. The admin sees a list of conflicts such as:

- Double-booked slots
- Overlapping schedules

39. The admin can manually resolve conflicts by reassigning patients.

40. Urgent escalations (e.g., patient in crisis) can override system rules to prioritize or re-route the appointment.

8.8. Notification & Communication Management

8.8.1. Notification & Broadcast Management

41. The admin can create, schedule, and send system-wide or targeted notifications to:

- Patients
- Doctors
- Hospital Admins

42. The portal supports read receipts.

43. Segmentation by region, speciality, or user type

44. Push notifications and SMS can be configured for critical alerts.

8.8.2. Chat & Communication Logs

45. The admin can monitor or moderate chat conversations if flagged.

46. The portal stores chat transcripts for a set duration (configurable) and provides search functionality by:

- Patient name
- Doctor name
- Date
- Keywords

47. The system ensures compliance with dispute resolution.

8.9. System & Data Management

8.9.1. Admin & Role Management

48. The master TeleHealthAdmin can create new admin roles (e.g., Finance Admin, Hospital Onboarding Admin)

49. The admin can assign granular permissions to control access to specific pages/features.

50. Role-based access ensures data privacy and compliance.

8.9.2. Data Export & Backup Management

51. The admin can export data such as:

- Appointments
- Revenue
- User details

52. Supported formats include:

- CSV
- Excel
- PDF

53. The portal provides:

- Backup scheduling (daily or weekly backups)
- Logs of successful/failed backup attempts

8.10. Hospital-Specific Admin Assignment

8.10.1. Assigning Hospital Admins

54. The master TeleHealthAdmin can assign a dedicated admin for a specific hospital, granting them permissions to manage:

- Hospital Data: Update hospital details, services, and pricing.
- Doctors: Add, edit, or remove doctors associated with the hospital.

55. The system ensures that assigned admins have restricted access based on their role, including:

- Hospital-Specific Access: Admins can only manage data for their assigned hospital.
- Doctor Management: Admins can update doctor details, availability, and consultation charges.

8.10.2. Permissions & Access Control

56. The system allows the master TeleHealthAdmin to define granular permissions for assigned admins, including:

- View-Only Access: For financial or operational data.
- Edit Access: For updating hospital or doctor details.

57. The system ensures that all actions performed by assigned admins are logged for auditing purposes, including:

- Action Type: Add, edit, or remove.
- User: Name and role of the admin acting.
- Date & Time: When the action was performed.

8.10.3. Real-Time Updates

58. The system ensures that changes made by assigned admins are reflected in real-time across the platform, including:

- Hospital Details: Updated information is displayed in the user app.
- Doctor Availability: Updated schedules visible to patients.

59. The system sends notifications to the master TeleHealthAdmin for critical updates, such as:

- New Doctor Additions: Notification when a new doctor is added.
- Price Changes: Notification when consultation or service charges are updated.

8.11. Health Services & Package Management

8.11.1. Add, Edit, or Remove Health Services

60. The admin can add new health services to the platform, including:

- Hospital (Multi-Select)
- Service Name: Name of the service (e.g., "Blood Test," "Full Body Check-Up").
- Service Type: Category of the service (e.g., lab test, consultation, home care).
- Price: Original price of the service.
- Discount: Optional discount percentage or fixed amount.
- Description: Brief details about the service (e.g., tests included, duration).

61. The admin can edit existing health services to update:

- Service Name: Update the name of the service.
- Price: Adjust the original or discounted price.
- Description: Modify the details of the service.

62. The admin can remove health services that are no longer offered, ensuring they are no longer visible in the patient-facing marketplace.

8.11.2. Real-Time Updates

63. The system ensures that all updates (add, edit, remove) to health services and bundled offers are reflected in the patient-facing marketplace in real time.

64. The system sends notifications to the admin when:

- A new service or bundle is successfully added.
- An existing service or bundle is updated or removed.

8.12. Data Security & Compliance

8.12.1. HIPAA-Compliant Servers

- 65. The system ensures that all data is stored in HIPAA-compliant servers, providing a secure environment for sensitive patient and hospital information.
- 66. The system undergoes regular audits to ensure continuous compliance with HIPAA regulations.
- 67. The system provides detailed logs of data access and modifications for compliance reporting.

8.12.2. Encryption

- 68. The system encrypts all data with 256-bit encryption during transit and at rest, ensuring that sensitive information is protected from unauthorized access.
- 69. The system uses industry-standard encryption protocols (e.g., AES-256) to secure data.
- 70. The system ensures that encryption keys are securely managed and rotated periodically.

8.12.3. Two-Factor Authentication (2FA)

- 71. The system implements two-factor authentication (2FA) for all admin accounts, adding an extra layer of security to prevent unauthorized access.
- 72. The system supports multiple 2FA methods, including SMS, email, and authenticator apps.
- 73. The system allows admins to enforce 2FA for all users within their organization.

8.12.4. Role-Based Profiles

- 74. The system allows admins to set up role-based profiles for staff, ensuring that:
- 75. Doctors: Have full access to patient files and medical records.
- 76. Staff: Have restricted access, limited to clinic operations (e.g., scheduling, billing).
- 77. The system ensures that sensitive patient information is only accessible to authorized roles.

8.12.5. Data Backup

- 78. The system takes multiple backups of all data, stored in geographically distributed locations, to ensure data availability and recovery in case of:
- 79. Natural Disasters: Data remains safe even if one location is affected.
- 80. Technical Failures: Redundant backups ensure no data loss.
- 81. The system provides an option for admins to manually trigger backups or restore data from backups when needed.

8.12.6. Virus Protection

- 82. The system stores all data in the cloud, protecting it from local viruses or malware that may affect user devices.
- 83. The system ensures that data integrity is maintained, with no risk of corruption from local threats.

Health Services System

9. Health Services System Requirements Document

9.1. Patient App (Web and Mobile App)

9.1.1 Authentication & Profile Setup

1. The app allows new users to register using their mobile number and email ID.
2. The app sends an OTP (One-Time Password) to the provided mobile number or email for verification.
3. The app ensures that the account is secure and verified before granting access.
4. The app supports facial recognition as an alternative login method for registered users.
5. The user can enable facial recognition in their profile settings for quick and secure access.

9.1.2 Profile Setup

6. The app allows users to fill in their profile with the following details:
 - First Name.
 - Last Name.
 - Marital Status.
 - Age.
 - Blood Group.
 - Existing Health Conditions.
 - Allergies.
 - Current Medications.
7. The app uses this information to personalize health service recommendations.
8. The app prompts users with an extended health questionnaire upon first login, including:
 - Weight.

- Height.
- Chronic Conditions.
- Recent Symptoms.
- Lifestyle Habits (diet, exercise, smoking, alcohol).

9. The app uses the questionnaire responses to tailor offers and recommendations to the user's needs.

9.1.3 Health Services Marketplace

Regional Selection

10. The app allows users to select their emirate, region, or location on a map during onboarding.
11. The app prompts users to grant location access for accurate service filtering.
12. The app displays health check-up services and offers available in the selected region.

Service Listing & Details

13. The app displays a marketplace of hospitals, clinics, and medical centres offering health check-up packages.
14. The app highlights exclusive discounts (e.g., 15% off) to motivate users to book through the platform.
15. The app provides detailed information for each service/package when selected, including:
 - Hospital/Clinic Name.
 - Health Check-Up Packages.
 - Pricing: Original and discounted prices (with 15% off clearly shown).
16. The user can select a test to view the full details page of that package including the following details:
 - Tests Included: List of tests or services included in the package.

- **Service Timings:** Available time slots for booking.

17. The app allows users to select an available date and time from an integrated calendar.

18. The app displays a summary of the selected service, date, and time before adding it to the cart.

Add to Cart

19. The app allows users to add health check-up packages to their cart.

20. The app displays the discounted price (with 15% off) in the cart.

Checkout & Payment

21. The app provides a secure checkout process for users to complete payment.

22. The app ensures that the discounted price is applied during payment.

23. The app sends a confirmation notification (email/SMS) once the booking is successfully paid for.

9.1.4. Booking & Payment Flow

Appointment Booking

24. The app sends immediate confirmation of health check-up bookings via SMS and email.

25. The app displays appointment details in the user's dashboard, including:

- **Appointment Date & Time:** Scheduled date and time of the health check-up.
- **Service Details:** Name of the health check-up package or tests booked.
- **Result Availability:** Link to view test results when available.

26. The app organizes booked appointments in the Appointment Section of the user's dashboard.
27. The app allows users to view details of each appointment, including:
- Hospital/Clinic Name: Name of the facility where the appointment is scheduled.
 - Service Type: Type of health check-up or test booked.
 - Status: Confirmed, completed, or cancelled.

Test Results & Medical Records

28. The app displays a history of all medical tests and health check-ups booked by the user, including:
- Test Name: Name of the test or health check-up package.
 - Date & Time: When the test was conducted.
 - Hospital/Clinic Name: Facility where the test was performed.
 - Cost: Amount paid for the test or package.
29. The app allows users to view, download, and share test reports in PDF format.
30. The app provides an option to email test reports directly from the app.

User Notifications & Reminders

31. The app sends timely reminders for upcoming health check-up appointments, including:
- 4 Hours Before: Reminder to prepare for the appointment.
 - 1 Hour Before: Reminder to leave for the appointment.
 - 15 Minutes Before: Final reminder to attend the appointment.
32. The app sends a notification when test results are ready, including:
- Test Name: Name of the test or health check-up.
 - Result Availability: Link to view or download the results.
33. The app ensures notifications are clear and in good English.

9.1.5. Footer & Navigation

Global Navigation

34. The app provides a footer with clear navigation links to the following main sections:

- Home: Return to the main dashboard.
- Orders/Records: View past orders, appointments, and medical records.
- Offers: Browse exclusive discounts and promotions.
- Cart: Review and manage items added to the cart.
- Account: Access profile settings, login details, and preferences.

35. The app ensures the footer is visible on all pages for easy navigation.

Cart Functionality

36. The app allows users to add health check-up packages or services to the cart.

37. The app displays a summary of items in the cart, including:

- Service Name: Name of the health check-up package or service.
- Price: Original and discounted prices (if applicable).
- Date & Time: Selected appointment slot (if applicable).

38. The app retains items in the cart even if the user logs out and logs back in.

39. The app allows users to review and modify cart items before proceeding to payment.

9.2. Hospital & Partner Portal (Web)

9.2.1 Appointment Management

40. The system allows hospital admins to log into their portal and view incoming appointment bookings, including:

- Patient Name: Name of the patient who booked the appointment.
- Appointment Date & Time: Scheduled date and time of the appointment.
- Service Type: Type of health check-up or consultation booked.
- Status: Pending, confirmed, or cancelled.

41. The hospital can acknowledge new bookings within a set timeframe, ensuring the system sends a final confirmation to the patient.

42. The system displays incoming appointment bookings on the admin portal, including:

- Patient Details: Name, contact information, and health check-up package.
- Appointment Details: Date, time, and service type.

43. The hospital can acknowledge bookings, triggering a confirmation notification to the patient via SMS and email.

9.2.2 Booking Calendar Integration

44. The system integrates with the hospital's booking calendar to display real-time slot availability to users.

45. The system prevents duplicate bookings by blocking slots temporarily (5-10 minutes) when a user adds a service to their cart.

46. If a parallel booking occurs, the system fails the payment process and prompts the user to select a different time or date.

9.2.3 Price & Package Management

47. The system allows hospitals to update service prices, packages, and available slots directly from the portal.

48. The system ensures that changes are reflected immediately in the user-facing app, including:

- Service Prices: Updated pricing for consultations, health check-ups, and other services.
- Package Details: Changes to included tests or services.
- Slot Availability: Updated time slots for bookings.

9.2.4 Report Generation

49. The system allows the hospital to generate reports on:

- Number of Tests Conducted: Total tests performed over a selected timeframe.
- Revenue Earned: Total revenue generated from consultations, tests, and other services.
- Demographics: Breakdown of patients by age, gender, and region.

50. The system provides filters for generating reports by:

- Timeframe: Daily, weekly, monthly, or custom date range.
- Service Type: Filter by consultations, tests, or other services.
- Doctor: View performance metrics for individual doctors.

9.3. Admin Dashboard (Web)

9.3.1 Commission & Price Control

51. The system allows the admin to modify commission rates dynamically for each hospital or service, including:

- Commission Percentage: Adjust rates (e.g., from 10% to 15%).
- Effective Immediately: Changes are reflected in real-time across the platform.

52. The system ensures that updated commission rates are applied to all transactions, including:

- Consultations: Commission on consultation fees.
- Health Check-Ups: Commission on test packages.
- Other Services: Commission on additional services (e.g., nurse-at-home).

9.3.2 Partner Management

53. The system allows the admin to add, remove, or update hospital and service listings, including:

- Hospital Details: Name, location, specialities, and contact information.
- Service Listings: Health check-up packages, consultations, and other services.

54. The system ensures that changes are immediately reflected in the user-facing app, ensuring only up-to-date offers are displayed.

9.3.3 Comprehensive Reporting

55. The system allows the admin to generate detailed reports on platform performance, including:

- Sales: Total revenue generated from consultations, tests, and other services.
- Commission: Total commission earned by the platform.

- Frequently Booked Tests: Most popular health check-up packages or tests.
- User Demographics: Breakdown of users by age, gender, and region.

56. The system provides filters for generating reports by:

- Timeframe: Daily, weekly, monthly, or custom date range.
- Hospital/Service: Filter by specific hospitals or services.
- Region: View performance metrics by emirate or region.

10. Conclusion

The DoctorNow Teleconsultation Platform redefines digital healthcare by connecting patients, doctors, and administrators through a secure, user-centric ecosystem. By automating workflows, enforcing compliance, and providing actionable insights, DoctorNow reduces administrative overhead, minimizes errors, and enhances patient outcomes. Its scalable architecture ensures adaptability for single clinics or multi-branch hospital chains, while robust security measures safeguard sensitive data. DoctorNow is not just a tool but a transformative solution for modern, efficient, and equitable healthcare delivery.

11. Feedback Points

4. Patient Dashboard Requirements

4.1. User Authentication & Profile Setup

4.1.1. User Authentication

1. Mobile number is mandatory for user registration.
2. During registration, an OTP will be sent for confirmation.
3. Users can opt for auto-pick OTP functionality to populate the OTP fields automatically.
4. Upon successful OTP verification, a confirmation message should be displayed, and the user should be redirected to the password setup page.
5. After setting a password, the user will proceed to profile setup. A “Skip” option should be provided at the top of the page.
6. If the user skips the profile setup, they should still be able to log in, navigate through the app, and view doctors, appointments, and health services. However, when booking a consultation, all necessary details must be filled in before proceeding.

4.1.2. Profile Management

7. The user can complete their profile with the following mandatory fields: First Name, Last Name, Age, Gender, Emirates ID, Nationality, and Date of Birth.
8. Upon completing these fields, user registration should be marked as successful.
9. Personal Details: Email ID and Emirates ID should be mandatory for profile setup.

10. For multiple family profiles, Emirates ID must be unique, and validation should be implemented.
11. Validation should be in place for the Date of Birth and Emirates ID. The Emirates ID format should be 784-YYYY-XXXXXXX, where:
 - The 4th to 7th digits represent the date of birth.
 - The first three digits are always 784.
12. The system should prompt users to complete their remaining health information:
 - Insurance Details (Optional): Insurance Company, Insurance Plan, and Insurance Number.
 - Initially, these fields will be open text fields. Future releases will integrate API-based dropdowns.
 - Health Information: Existing conditions, allergies, current medications, and lifestyle habits.
 - Additional optional details include Weight, Height, Blood Pressure, Sugar Levels, Chronic Conditions, Recent Symptoms, and Health Goals.

4.2. Home Page

13. The Home Page should have two sections: one for health services and another for consultations. These sections can be displayed separately or combined; the final design will be decided in the UI phase.
14. A small horizontal rectangular placeholder should be allocated for advertisements. These ads should be controlled and updated in real-time from the hospital admin panel.

Specialty Section:

15. Users can view medical specialties displayed as icons on the homepage.
16. A horizontal scroll/swipe feature should allow users to navigate through specialties, with a “View All” button to access a detailed speciality list.
17. Selecting a speciality should display a list of doctors in that category, sorted based on locality.
18. A central search bar should be available on top, allowing users to search for Doctors, Specialties, Facilities, and Locations with filtering and sorting options:
19. Sorting Options: Distance (Nearest First), Name (A-Z).
20. Filter Options:
 - Consultation Date
 - Doctor Gender
 - Facility (Search & List)
 - Practitioner Spoken Language
 - Distance Range
 - Country of Education
 - Country of Professional Experience
 - Consultation Fee
 - Emirate
21. Search Functionality:
 - If the user searches for a doctor’s name, results should display the doctor after entering at least three characters. If multiple matches exist, they should be listed.
 - If the user searches for a speciality, all doctors related to that speciality should be displayed.
 - If the user searches for a facility name, all doctors associated with that facility should be displayed.

- If the user searches for a location, doctors from facilities within that location should be listed.

Hospital Section:

22. Displays a list of up to 20 nearby hospitals/clinics based on user location.
23. Users can swipe horizontally to view more facilities or click “View All” to see the complete list.
24. The hospital List Page should have sorting options for Distance (Nearest First) and Name (A-Z).
25. A search bar for “Facility” and “Location” should be present at the top.
26. Selecting a hospital/facility/clinic name should display facility details, followed by the list of specialities available at that facility and the associated doctors.
27. The speciality section on the hospital details page should be interactive, allowing users to filter doctors by speciality.
28. Doctor Tiles: Each doctor listing should include:
 - Doctor Name
 - Doctor Photo
 - Specialization
 - Languages Spoken
 - Clinic/Facility Name
 - Distance from the patient’s location
 - Next Available Consultation Slot (Date & Time)
 - A “View More” option
 - A “Save” option to add the doctor to a favourites list
29. Doctor Details Page: Clicking on a doctor should display:
 - Doctor Photo & Name
 - Specialization

- Languages Spoken
- Available Time Slots (with an option to view more)
- Qualifications
- Services Provided

30. Validation:

- Children (patients under 18) should only be allowed to book appointments with pediatric doctors.
- Adults should not be able to book appointments with pediatric specialists.

31. All doctor details should be dynamically updated based on entries from the Hospital Admin Portal or App Admin Portal.

32. Any newly added doctor should be reflected in real time or within 30 minutes.

33. Doctor Addition Process: Should follow a Maker-Checker Process for approval.

Health Services Section:

34. The Home Page should display a list of available health services. Users can swipe horizontally to browse or click “View All” to see the complete list.

35. Selecting a service should display all providers offering that service.

36. Sorting & Filtering Options:

- Sorting: Distance (Nearest First), Name (A-Z).
- Filters: Consultation Date, Facility, Distance Range, Price, Location, Emirate.

37. **Service Tiles:** Each service listing should include:

- Facility Logo/Image
- Package Name
- Price

- Discounted Price (if applicable, with the original price struck through)
- Offer Expiry Countdown (Optional)
- A “View Details” button

38. **Discount Logic:** Discounts are derived from two values:

- Facility/Hospital Discount %
- App Discount %

39. Both values are entered via the Hospital Admin Portal and App Admin Portal.

40. Any app-admin-based discounts require approval from the Super App Admin.

41. Say for example service or consultation is 1000AED, a 10% DISCOUNT IS GIVEN BY the Health provider and the app admin shares a 5% discount..total 15% discount should be shown in the patient UI 10% discount to be added by the hospital admin, and checker approval required by app admin app.

42. Health Service Booking Process:

- Displays Health Centre Description, Price, Savings, Offer Expiry, Package Details, Instructions, and Available Booking Slots.
- Users can confirm bookings and proceed to payment.

4.6. Appointment Booking

43. Before proceeding with a booking, users must provide Emirates ID details, including the front and back copies and the ID's expiry date. The system should not allow users to proceed without this information.

44. Users can book both consultations and health services through the platform.

45. A confirmation page should display booking details, payment amount, and any applicable discounts. Users should be able to select a payment provider and proceed to the payment gateway.
46. Upon entering payment details and successfully confirming the payment via OTP, a confirmation screen should appear, displaying:
- Unique booking reference number
 - Appointment time slot
 - Payment details
47. An invoice with booking and payment details should be sent to the user's registered email address. A confirmation message should be sent to the user's mobile number.
48. Users should have an option to add the booking to their calendar for reminders.
49. All remaining notifications and features should be implemented as per the provided Excel sheet.

4.11. Consultation

Doctor Face Detection:

50. Once the doctor joins the session, the system should verify their identity using face recognition technology.
51. If a proxy is detected, an alert notification should be sent to the hospital admin for immediate action.
52. The issue should be resolved within 5 minutes. If not, an apology message should be shown to the patient, along with an option to either reschedule the appointment or request a refund.
53. Facial detection accuracy should be 100%, as false alarms can disrupt the consultation process.

4.12. Post-Consultation Process

- 54. A notification should confirm consultation completion.
- 55. Patients should receive an in-app message stating that their prescription will be available under "My Appointments → Consultation Details" within 15 minutes for medication purchase.
- 56. If a prescription is issued, it should appear under "My Appointments → Today's Consultations" within 10 minutes of consultation completion.
- 57. Pharmacy integration will be implemented in a future release.

4.13.2. Appointment Management

- 58. Users should be able to join upcoming consultations directly from the appointment card using a "Join Now" button.
- 59. The "Join Now" button should be enabled 15 minutes before the scheduled appointment time.

4.13.3. Prescription & Report Access

- 60. Users should be able to view and download prescriptions, consultation notes, and lab reports associated with past appointments.
- 61. The hospital admin should attach the reimbursement form within 1 hour of consultation completion.
- 62. Once uploaded in the hospital portal, the reimbursement form (with hospital seal) should be automatically reflected in the app under the respective patient appointment.

4.21. Family Profile Management

4.21.1. Multi-Profile Setup

- 63. Users should be able to add multiple family profiles under their account.
- 64. Each family profile must include the Emirates ID details (ID number, front and back image, and expiry date) before booking any consultation.
- 65. Insurance details (company, plan, and number) can be optionally added for each family member.
- 66. A disclaimer should be displayed stating that reimbursement will only be processed if valid insurance details are provided.

5.2. Dashboard

- 67. The dashboard should display pending notifications and alerts for the doctor.
- 68. If a patient joins the lobby (which opens 15 minutes before the consultation time), a distinct notification alert should be sent to the doctor's portal, differentiating it from other notifications.
- 69. Additional reminders and notifications for the doctor to join the consultation should be triggered as per the Excel specifications.

5.5.3. Cancellation Prompt

- 70. The doctor will receive a cancellation prompt if the patient does not join the consultation on time.
- 71. The system should trigger a confirmation prompt before cancelling the booking.
 - This prompt should be triggered 5 minutes after the doctor joins the session.

- If the doctor decides to cancel, the system should provide options to:
 - Reschedule the consultation
 - Send a cancellation note to the patient

72. If the patient fails to join, the system should handle the refund and no-show fee based on the patient's confirmation:

73. If the patient confirms the cancellation, the booking amount should be refunded after deducting any applicable no-show fees.

74. If the patient does not confirm, the hospital's policy on no-shows should determine the refund or penalty.

6.2. Dashboard Page

6.2.1. View Dashboard & Welcome Message

75. The dashboard header should include navigation links to:

- Dashboard
- Doctors
- Consultations
- Revenue
- Patient Search

6.3.1. Consultation Page Overview

76. The system should display a detailed list of consultations with the following statuses:

- Booked
- Completed
- Cancelled

6.5. Consultations Section

77. Upon confirmation of a consultation, notifications should be sent to both the patient and the doctor.

78. Additionally, an SMS or WhatsApp notification should be sent to the customer.

6.6. Patients Section

6.6.1. Patient List

79. The system should display a list of all patients associated with the hospital, including:

- Emirates ID details
- Insurance details

6.6.2. Patient Profile

80. The patient profile should include:

81. Uploaded Documents: Lab reports, prescriptions, and other medical records.

82. Users should have the option to upload reports immediately after a consultation.

83. When uploaded, documents should follow a proper naming convention in a password-protected format, e.g.:

- JAFE_BLOODTEST_DATE (EPOCH FORMAT)

6.6.4. Lab Reports & Documents

84. A placeholder should be added for the suggested tests prescribed by the doctor.

85. This feature should be incorporated into the complete workflow, covering:

- Patient consultation booking

- Hospital dashboard
- Admin dashboard

6.6.5. Patient Communication

86. The hospital admin should be able to view past communication with the patient, including:

- Messages: Chat history between the patient and doctor.
- This chat history should be accessible only to the doctor and patient.
- It should not be visible in the hospital admin dashboard or should be handled securely.

6.7.6. View & Manage Commission Details

87. Commission details should be editable, following a Maker-Checker process between the Hospital Admin and App Super Admin.

6.10. Real-Time Payment Update Tracker

6.10.1. Live Payment Dashboard

88. Failed payments should be handled under three different scenarios:

- Payment failed, booking failed, and no money was deducted.
- Payment failed, booking failed, money was deducted from the customer but credited to the app.
- Payment failed, booking failed, money was deducted from the customer but not credited to the app.

89. A tracking mechanism should be in place to monitor these scenarios.

6.12. Security & Role-Based Access Control

6.12.1. Sub-Account Creation

90. Sub-accounts should be created for role management, including doctors.

91. Note:

- Health service package management (addition, editing, package cost, discount, commission %) is not covered in this BRD.
- For reference, please refer to the Health Services section in the Patient Portal above and the Health Services Tab in the Excel.

7.6.2. Commission Rate Management

92. A Maker-Checker process should be implemented for commission rate updates.

93. The Hospital Admin should have the ability to:

- **Manage health services bookings.**
- **Edit and add new health services provided by the hospital.**
- **Onboard only medical centres for health services.**
- **Onboard hospitals for both health services and consultation doctors.**

7.12.6. Data Backup

94. The system should perform multiple backups of all data.

95. Data should be stored in geographically distributed locations within the UAE.

96. Backups should be maintained across multiple data centres for redundancy and security.

12. Annexure-Technical Touchpoints

(For Reference-For this MVP-Integrate with Nabidh for Dubai and Riyati for Sharjah and Northern emirates)

The 3 big health-data rails in the UAE

- **NABIDH** → Dubai's health information exchange (HIE) run by Dubai Health (DHA). It connects public/private providers in the Emirate of Dubai and standardizes clinical-data exchange (HL7v2/FHIR). Use this for Dubai facilities' clinical record sharing. (PMC)
- **Riyati (NUMR)** → The federal platform run by MOHAP; it's the national unified medical record and umbrella that is integrating with both NABIDH and Malaffi (so records can flow nationwide). Riyati also hosts the e-Claims Post Office for Northern Emirates (Sharjah, Ajman, UAQ, RAK, Fujairah). (Ministry of Health and Prevention - UAE)

In 2022 the authorities announced successful integration between Riyati, Malaffi, and NABIDH, enabling cross-emirate access to unified patient records. That's your backbone for national continuity of care. (Department of Health Abu Dhabi)

Insurance claims rails (where eligibility/claims actually move)

- **Dubai: eClaimLink (DHA)** is the official clearinghouse for electronic eligibility & claims between providers and payers/TPAs in Dubai. Your app (or the hospital PMS/EMR) submits eligibility (E1/E2) and claims here; eClaimLink routes to the correct insurer/TPA (e.g., NextCare, NAS, Neuron, Oman, ADNIC, etc.). It's not an insurer—it's the DHA exchange. (eclaimlink.ae)
- **Northern Emirates (incl. Sharjah):** Riyati e-Claims Post Office handles secure exchange of claims/eligibility between

providers and payers operating in those emirates. (Ministry of Health and Prevention - UAE)

MOHAP has publicly stated Riayati is now processing millions of insurance claims, underscoring that Northern Emirates claims traffic is on Riayati rather than eClaimLink. (Ministry of Health and Prevention - UAE)

So, for initial rollout (Dubai + Sharjah), Clinical HIE (records): Dubai facilities → NABIDH and Sharjah facilities → Riayati (NUMR) as the federal HIE layer. Cross-emirate continuity is achievable because Riayati ↔ NABIDH/Malaffi are integrated. (Department of Health Abu Dhabi)

Insurance (eligibility & claims):

Dubai claims → eClaimLink. (eclaimlink.ae)

Sharjah (Northern Emirates) claims → Riayati e-Claims Post Office. (Ministry of Health and Prevention - UAE)

eClaimLink is the DHA exchange, not an insurance company. Multiple payers/TPAs (NextCare, NAS, Neuron, etc.) connect to it; your eligibility and claims are routed to the right payer via eClaimLink. (In Abu Dhabi it's Shafafiya; in Northern Emirates it's Riayati e-Claims). (eclaimlink.ae)

Insurer/TPA landscape: DHA maintains a register of permitted insurers/TPAs for Dubai (you'll see NextCare, NAS, etc.). Your system doesn't integrate one-by-one with each payer; you integrate once with the government exchange, which fans out to all permitted payers. (Isahd)

Compliance with DHA/MOHAP Telehealth Standards and Integration Requirements

DHA standards require obtaining explicit patient **consent for telehealth** and maintaining confidentiality in all virtual consultations

DHA policy also requires compliance with the Dubai Health Insurance Corporation (DHIC) processes for telehealth approvals, e-claims, and reimbursement and **Riayati** for sharjah

The Federal ICT Health Law (No. 2 of 2019) and MOHAP guidelines mandate that patient health data must be stored within the UAE DHA explicitly prohibits storing or processing health records on servers outside the country (unless an exceptional MOHAP exemption is obtained) Additionally, UAE's **Personal Data Protection Law (PDPL)** imposes strict rules on handling health data (purpose limitation, consent, etc.) Your architecture must ensure all sensitive patient data (consultation recordings, EHR info, prescriptions) is **hosted on UAE-based servers/cloud** and complies with these local **regulations**

Medical records typically must be retained for at least **25 years** in UAE healthcare facilities

Ensuring **end-to-end encryption** of consultations and stored data (as you've noted for HIPAA) is necessary but not sufficient – you should also implement the UAE's **NESA** cybersecurity standards and any guidance from the Dubai Electronic Security Centre

DHA policy requires telehealth providers to **use approved, secure communication channels** for video consultations

Ensure your architecture uses a **TDRA-compliant telemedicine video platform** or obtains any needed clearances. Additionally, the platform should implement **audit logs and access controls** in line with DHA's information governance policies

DHA expects telehealth providers to report key performance indicators (KPIs) – such as number of teleconsultations, patient satisfaction, outcomes, etc

Build an analytics module to collect required KPIs (e.g. consultation counts,

average wait time, technical success rate, etc.) and generate reports compliant with DHA's format. By including a **Telehealth KPI reporting** feature, you'll meet DHA's performance monitoring compliance (as per **DHA/HRS Telehealth KPIs Guidelines** [dha.gov.ae](https://dha.gov.ae/dha.gov.ae)). This will likely become part of the DHA inspections or license renewals, so it's important to plan for it in your system design.

The Dubai Health Authority **mandates that all licensed health facilities in Dubai connect to the NABIDH HIE** to share patient records emrmagazine.com. Practically, this means any encounter (including a teleconsultation) should be reflected in the unified patient record. If your telehealth platform is operating across multiple hospitals/clinics, you need a strategy to ensure teleconsultation data (consult notes, diagnoses, e-prescriptions, etc.) are **uploaded to NABIDH**. Usually, this is achieved by having the encounter documented in the hospital's EMR which then sends data to NABIDH. If your platform functions as an extension of the hospital, it must at minimum **provide all required data to the hospital's EMR or directly to NABIDH** in the proper format. DHA specifies a minimum data set (patient demographics, vitals, problems, medications, insurance info, etc.) that must be captured and transmitted for each encounter remedico.app

Aligning with NABIDH also means capturing clinician identifiers (e.g. the doctor's DHA license/Sheryan ID) and insurance details in structured form, as these are part of the HIE record. In summary, you will likely need to **enhance your architecture to push teleconsultation records to each facility's EMR or an integration engine that sends data to NABIDH**.

Architecture-Reference only

Modular Platform Layers: Structurally, think of the system in layers: **Clients**, **Integration/API layer**, **Business Logic**, and **Data stores/External systems**. The clients are your patient mobile app, doctor web app, and admin portal (as in the BRD). These connect via the internet to the cloud backend. The backend should be designed as a set of **microservices or modules** – for example: Authentication service, Scheduling/Appointments service, Consultation/Video service, Payment service, **Insurance Integration service**, **EMR/HIE Integration service**.

EMR Integration (Hospital Systems): Unlike a standalone app, a compliant telehealth system in UAE should communicate with each hospital's **Electronic Medical Record (EMR)** system. In practice, there are a few ways to do this:

- **Direct API Integration:** If the target EMRs (Cerner, Epic, Insta, etc.) provide APIs or FHIR endpoints, the platform's integration layer can call those APIs to **pull or push data**. For example, when a patient selects a hospital and doctor, the app could fetch that doctor's schedule from the hospital's EMR (if available) to avoid double-booking and to reflect real-time availability. Similarly, after a teleconsultation, the platform could push a summary note or prescription back into the hospital's EMR for record-keeping.
- **HL7 Interface Engine:** Many EMRs (especially older or enterprise ones like Cerner/Epic) support HL7 messaging. You might deploy an **integration engine** (such as Mirth Connect or InterSystems Health Connect) as a middleware that translates between your platform's data and the EMR's HL7 messages. For example, when a teleconsult is completed, your system could send an **ADT message** or a **clinical document (MDM message)** via this engine to the EMR, which in turn forwards it to NABIDHdreamsoft4u.comdreamsoft4u.com.

